EE 3072
System Dynamics and Control

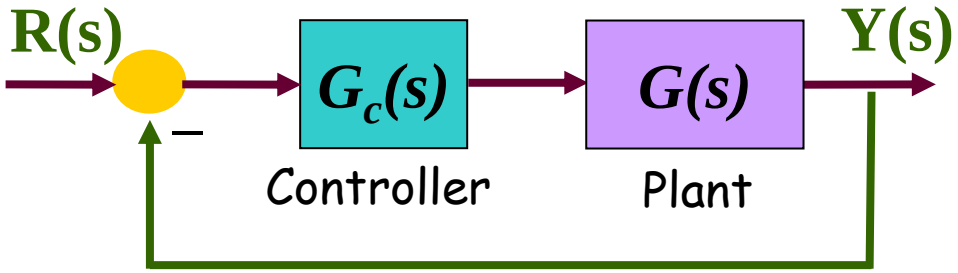
Control Systems Design by the Root-locus Technique

Outline

- Introduction
- Lead Compensator Design using Root-locus
- Lag Compensator Design using Root-locus
- Lead-Lag Compensator Design using Root-locus
- Possible Analog Implementation of the Compensators

Introduction

Consider the feedback control system



The plant is generally fixed and cannot be modified.

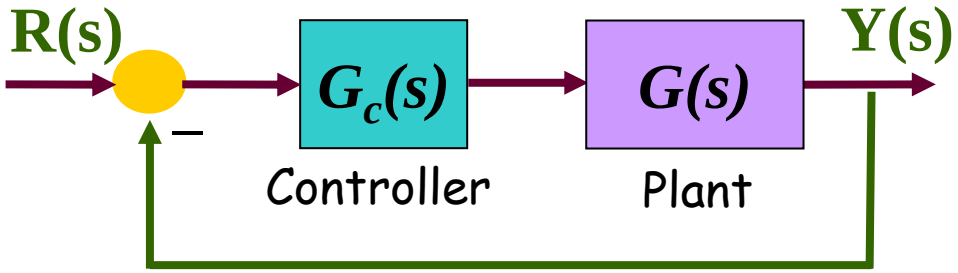
A controller must be designed so that the CL performance specifications are met.

The controller $G_c(s)$ is sometimes called compensator, since it is designed to compensate the deficit performance of the original system.

There are various methods for designing compensators, one of which is the root-locus method.

Introduction

Consider the feedback control system



The root locus method is a powerful method, which can be used for control systems design.

This is true especially when the desired specifications are given in terms of time domain requirements, such as the damping ratio and the undamped natural frequency of the dominant poles of the CL system.

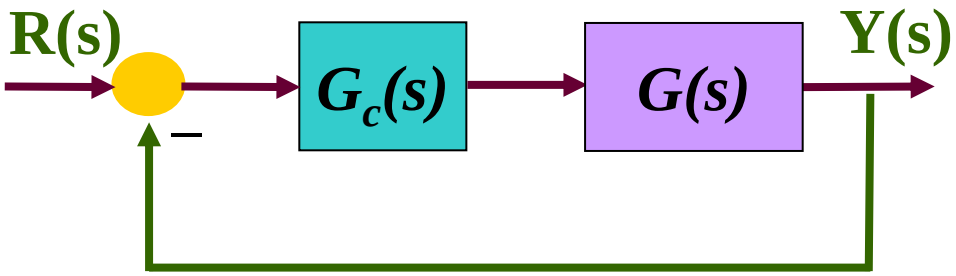
Various compensators are possible. We will consider

- Lead
- Lag
- Lead-Lag

compensators.

Lead Compensator Design Using Root-locus

The Lead Compensator



The TF of a phase lead compensator is given by

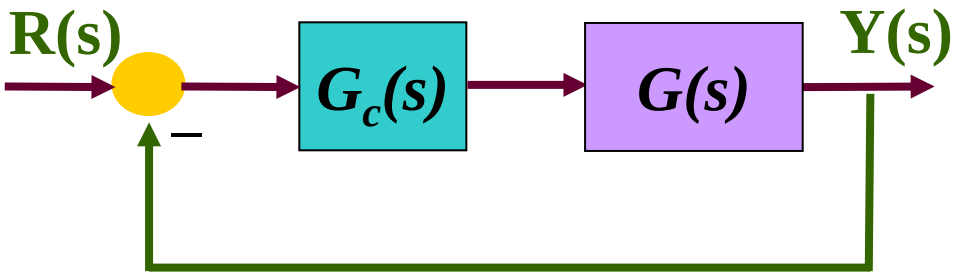
$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\alpha\tau}} = K_c \frac{s+z}{s+p}, 0 < \alpha < 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\alpha\tau$$

Note that since $0 < \alpha < 1$ we have $-p = -1/\alpha\tau < -1/\tau = -z$

The Lead Compensator



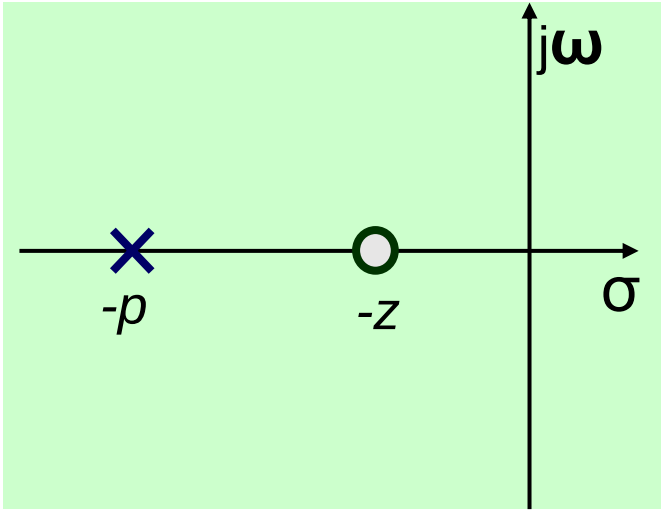
$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\alpha\tau}} = K_c \frac{s+z}{s+p}, 0 < \alpha < 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\alpha\tau$$

$$-p = -1/\alpha\tau < -1/\tau = -z$$

s-plane pole-zero configuration of a <u>lead compensator</u>

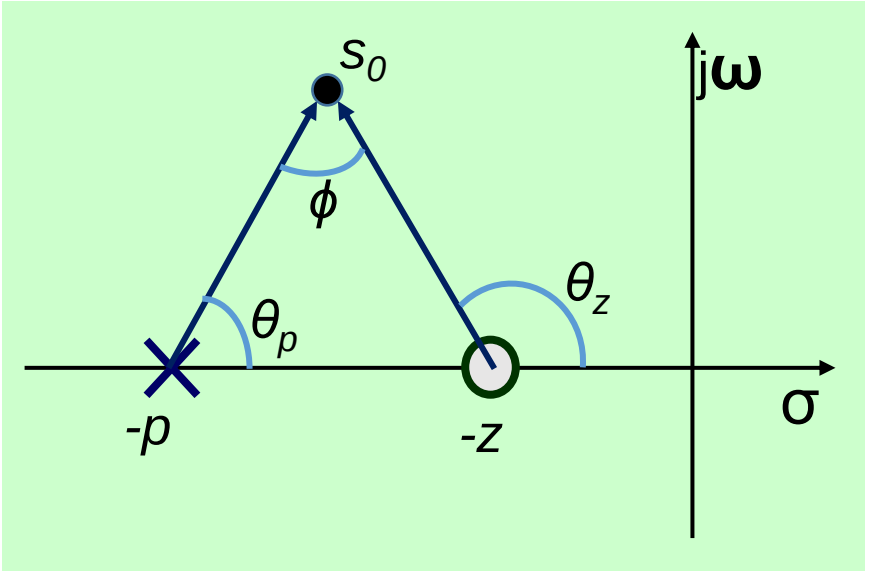


The Lead Compensator

Consider an arbitrary test point s_0 in the plane.

At $s = s_0$ we have

$$G_c(s_0) = K_c \frac{s_0 + z}{s_0 + p}$$



The magnitude of $G_c(s_0)$ is

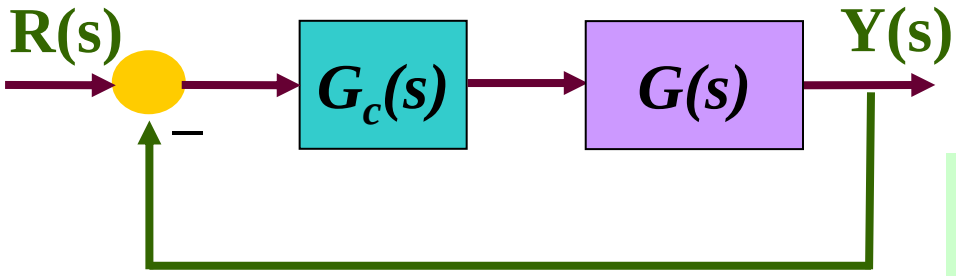
$$\left|G_c(s_0)\right| = K_c \frac{|s_0 + z|}{|s_0 + p|}$$

The angle of $G_c(s_0)$ is

$$\measuredangle\left(G_c(s_0)\right) = \measuredangle\left(s_0 + z\right) - \measuredangle\left(s_0 + p\right) = \theta_z - \theta_p = \phi$$

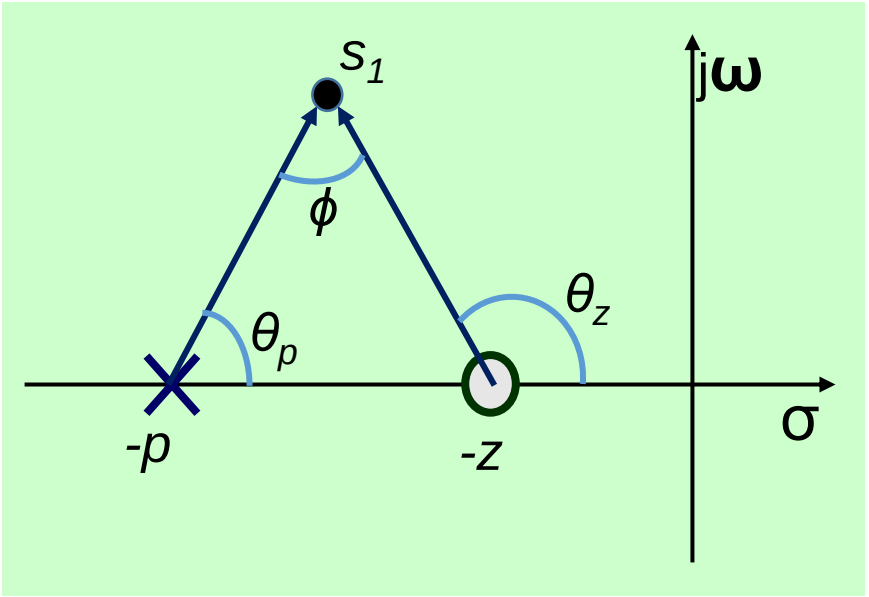
These are the values $G_c(s)$ will contribute to the OL TF $G_c(s)G(s)$ at a point s_0.

The Lead Compensator



Given a fixed plant $G(s)$ and a desired closed loop pole s_1 in the plane.

It is possible to design a compensator $G_c(s)$ with appropriate gain K_c and angle ϕ so that s_1 is a pole of the compensated CL system.



$$G_c(s) = K_c \frac{s+z}{s+p}, 0 < \alpha < 1$$

Lead Compensator Design Procedure

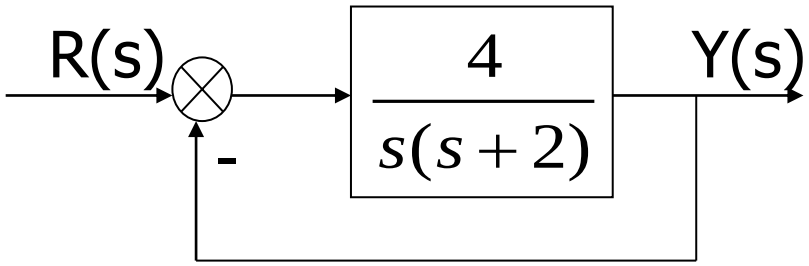
1. From the desired performance specifications determine the desired locations for the dominant closed-loop poles.
2. Draw the root-locus of the original system and determine whether it passes through the desired pole locations. If not, calculate the angle deficiency. This angle must be contributed by the compensator.
3. Place the zero of the compensator directly below the desired root locations (or to the left of the first two real poles, if any).
4. Determine the location of the pole of the compensator so that it contributes the necessary angle.
5. Determine the compensator gain from the magnitude condition.

Lead Compensator Design Procedure: Comments

- The discussed design procedure cannot directly address desired steady state error specifications (if any).
- ➢ If there is a specification on the desired steady state error, then the final design obtained from the above procedure may not necessarily satisfy the error specification.
- Therefore, once the compensator has been designed, it is needed to check and see whether all design specifications have been met.
- If not, then the pole and the zero of the compensator must be adjusted (i.e., the compensator redesigned) until all specifications are met.
- If the angle deficiency is more than 90°, then the zero of the compensator need to be placed to the right of the desired poles to be able to provide the necessary angle.

Lead Compensator Design

Example: Consider the system



Open-loop TF: $G(s) = \dfrac{4}{s(s+2)}$

Closed-loop TF: $T(s) = \dfrac{4}{s^2 + 2s + 4}$

For the current system we have: $\omega_n = 2,\ \zeta = 0.5,\ s_{1,2} = -1 \pm j\sqrt{3}$

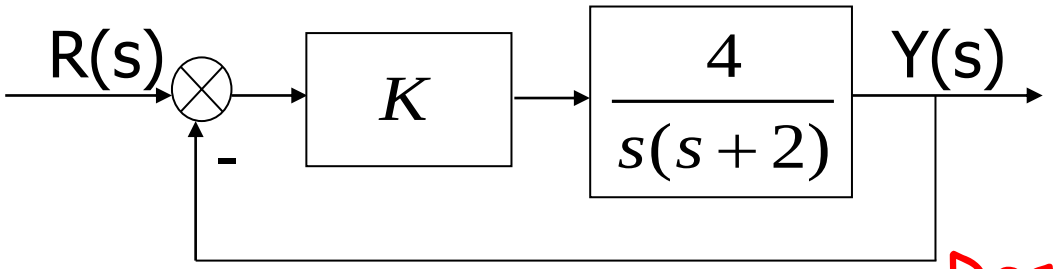
Design specifications:

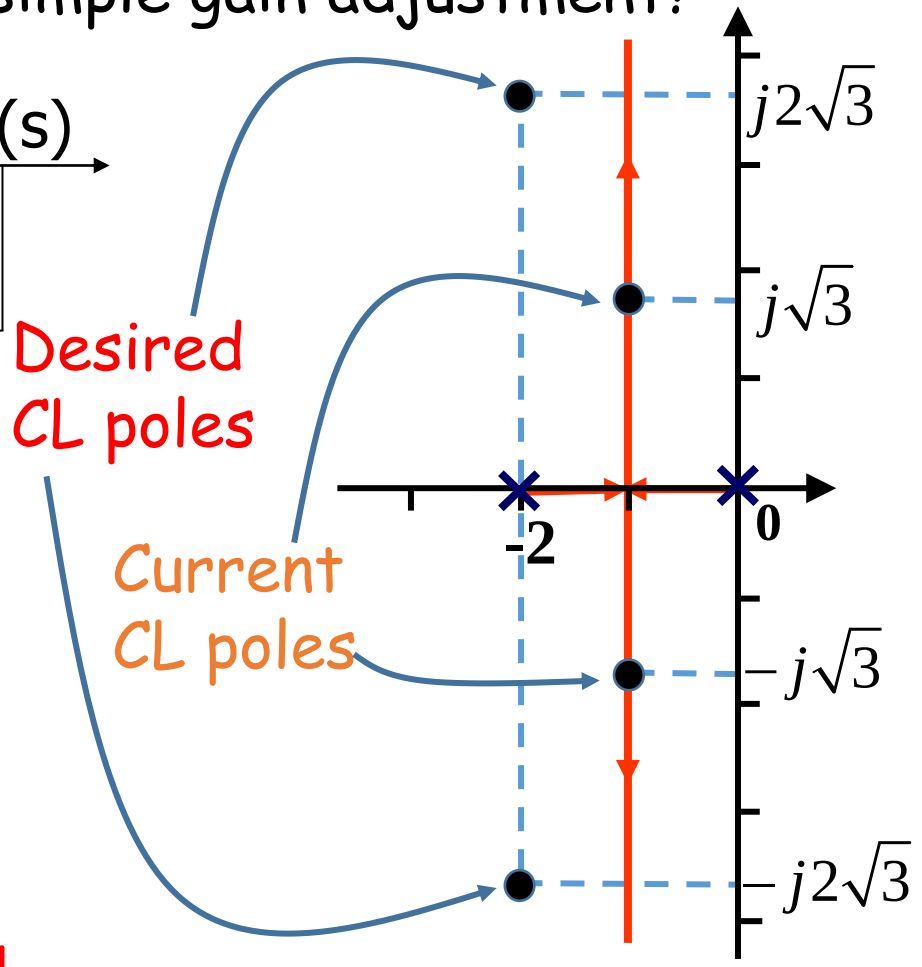
It is desired that the system has $\omega_n = 4,\ \zeta = 0.5$ for the dominant poles, and velocity error constant $K_v \geq 5$

The desired CL poles are located at: $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -2 \pm j2\sqrt{3}$

Lead Compensator Design

Can we solve the problem by a simple gain adjustment?





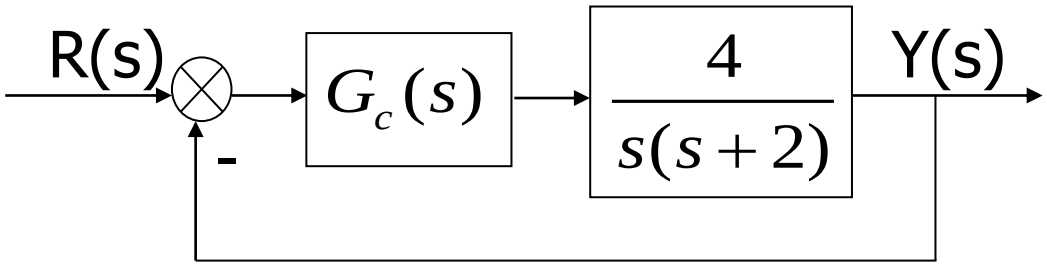
The desired CL poles do not lie on the root-locus.

Therefore, we cannot achieve them by simple adjustment of the gain.

We need to use a compensator!

Lead Compensator Design

Use a lead compensator



$$G_c(s) = K_c \frac{s + 1/\tau}{s + 1/\alpha\tau}, \qquad 0 < \alpha < 1$$

We must choose $G_c(s)$ such that all the design specifications are satisfied.

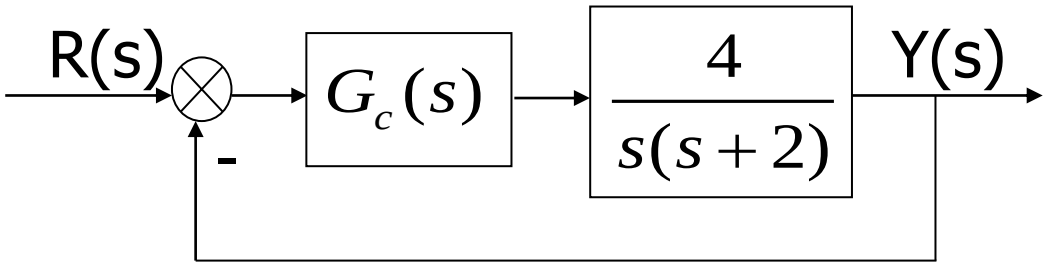
Calculate the angle of G(s) at the desired poles.

$$\measuredangle\left(G(s)\right)\Big|_{s=-2+j2\sqrt{3}} = \measuredangle\left(\frac{4}{(-2+j2\sqrt{3})(-2+j2\sqrt{3}+2)}\right)$$

$$= -\measuredangle\left(-2+j2\sqrt{3}\right) - \measuredangle\left(j2\sqrt{3}\right) = -120^o - 90^o = -210^o$$

Lead Compensator Design

Use a lead compensator



$$G_c(s) = K_c \frac{s + 1/\tau}{s + 1/\alpha\tau}, \qquad 0 < \alpha < 1$$

For the desired poles to be located on the locus we need

$$\measuredangle\left(G_c(s)G(s)\right)\big|_{s=-2+j2\sqrt{3}} = (2l+1)\times 180^o$$

$$\measuredangle\left(G_c(s)\right)\big|_{s=-2+j2\sqrt{3}} = 30^o$$

is needed

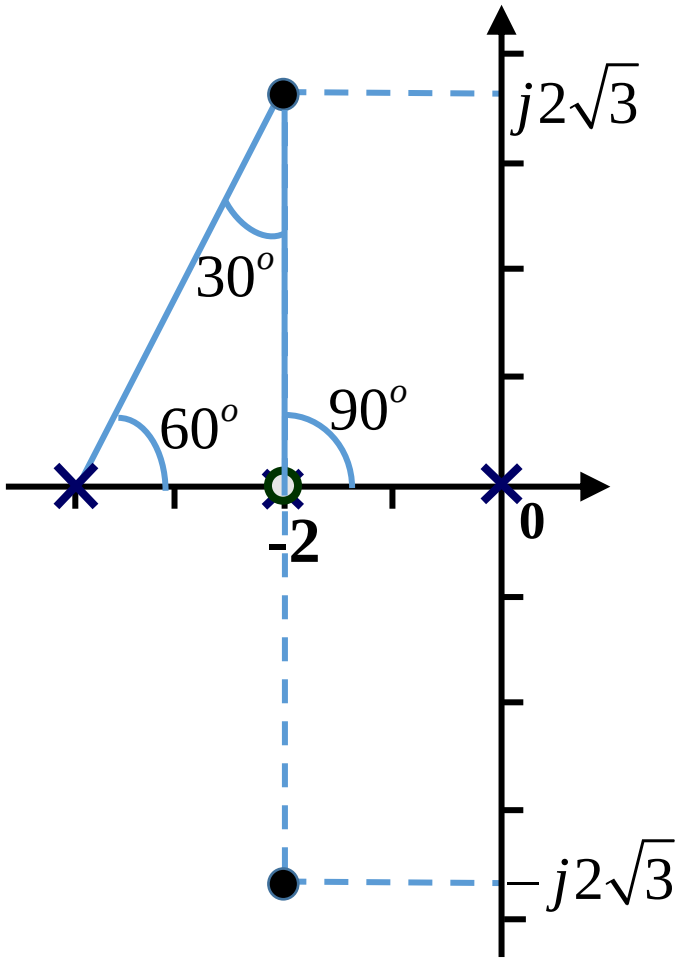
Lead Compensator Design

Using the design procedure choose

$$G_c(s) = K_c \frac{s+2}{s+4}$$

Its angle at the desired pole is

$$\measuredangle\left(G_c(s)\right)\big|_{s=-2+j2\sqrt{3}} = 90^o - 60^o = 30^o$$



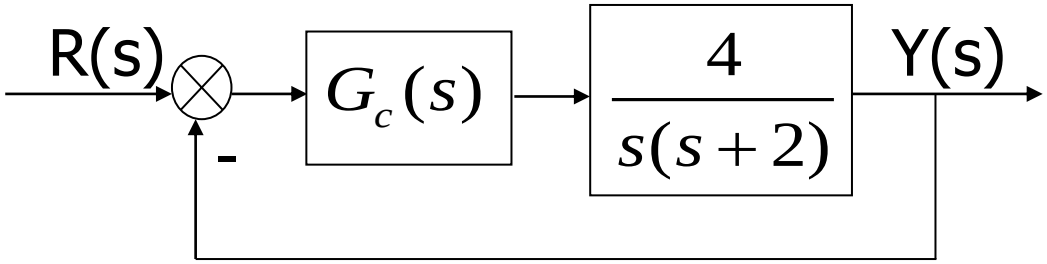
What is the value of K_c?

Using the magnitude condition

$$\left|\frac{4K_c(s+2)}{s(s+2)(s+4)}\right|_{s=-2+j2\sqrt{3}} = 1$$

$$4K_c = \left|-2+j2\sqrt{3}\right|\left|-2+j2\sqrt{3}+4\right| = 16 \quad \Rightarrow \quad K_c = 4$$

Lead Compensator Design



The compensator is

$$G_c(s) = 4\frac{s+2}{s+4}$$

The Closed-loop TF is $T(s) = \dfrac{16}{s^2+4s+16}$

For the compensated system we have

$$\omega_n = 4,\ \zeta = 0.5,\ s_{1,2} = -2 \pm j2\sqrt{3}$$

Natural frequency and damping ratio specifications are satisfied.

Check the velocity error constant specification.

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} s\frac{4(s+2)}{s+4}\frac{4}{s(s+2)} = 4 < 5$$

The steady state error specification is not satisfied!

We must redesign the compensator with new pole-zero configuration.

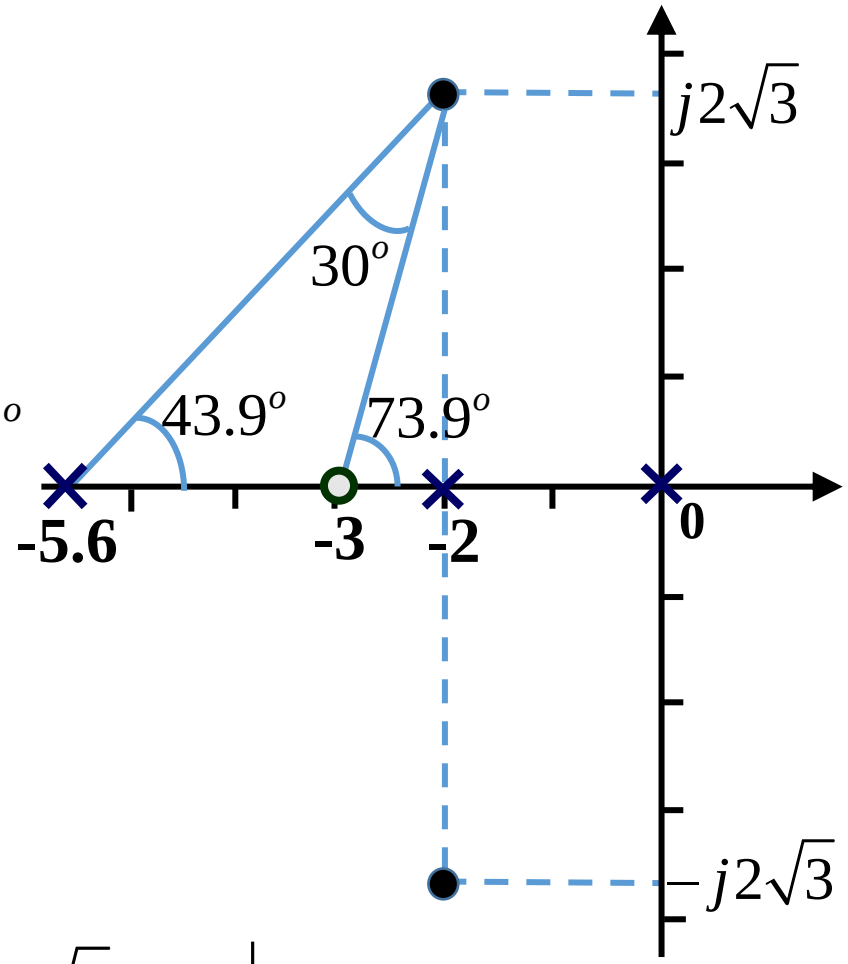
Lead Compensator Design

Choose the compensator

$$G_c(s) = K_c \frac{s+3}{s+5.6}$$

Its angle at the desired pole is

$$\measuredangle\left(G_c(s)\right)\Big|_{s=-2+j2\sqrt{3}} = 73.9^o - 43.9^o = 30^o$$



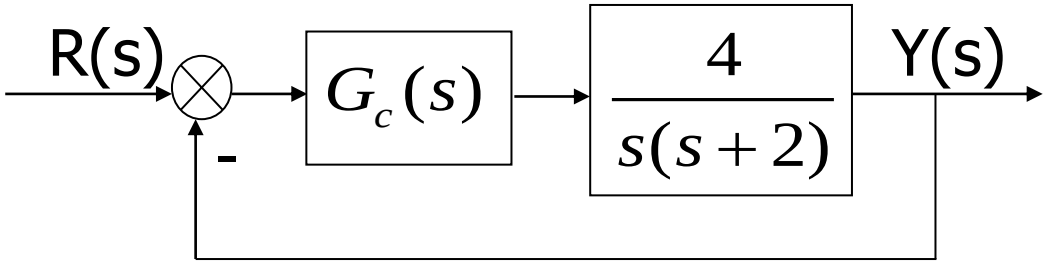
Using the magnitude condition

$$\left|\frac{4K_c(s+3)}{s(s+2)(s+5.6)}\right|_{s=-2+j2\sqrt{3}} = 1$$

$$4K_c = \frac{\left|-2+j2\sqrt{3}\right|\left|-2+j2\sqrt{3}+2\right|\left|-2+j2\sqrt{3}+5.6\right|}{\left|-2+j2\sqrt{3}+3\right|} = 19.20$$

$$K_c = 4.80$$

Lead Compensator Design



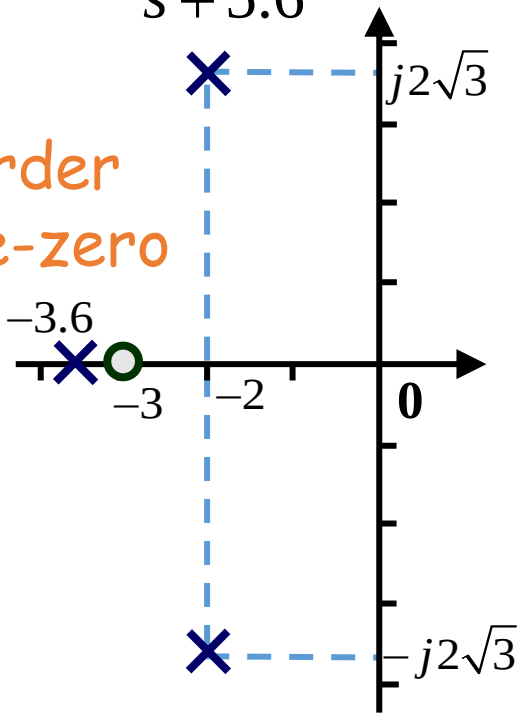
The compensator is

$$G_c(s) = 4.8\frac{s+3}{s+5.6}$$

The Closed-loop TF is

$$T(s) = \frac{19.2(s+3)}{(s+3.6)(s^2+4s+16)}$$

This is a third order system with pole-zero configuration

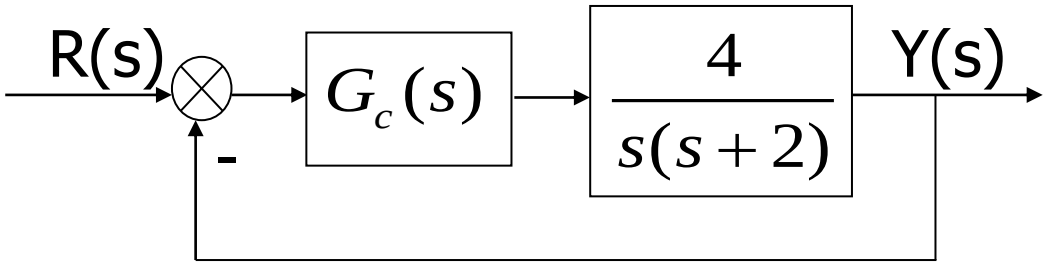


The poles at $s_{1,2} = -2 \pm j2\sqrt{3}$ are closest to the jw-axis – they are dominant poles

The compensator achieves the velocity error specification.

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} s\frac{4.8(s+3)}{s+5.6}\frac{4}{s(s+2)} = \frac{4.8\times 3\times 4}{5.6\times 2} = 5.14 > 5$$

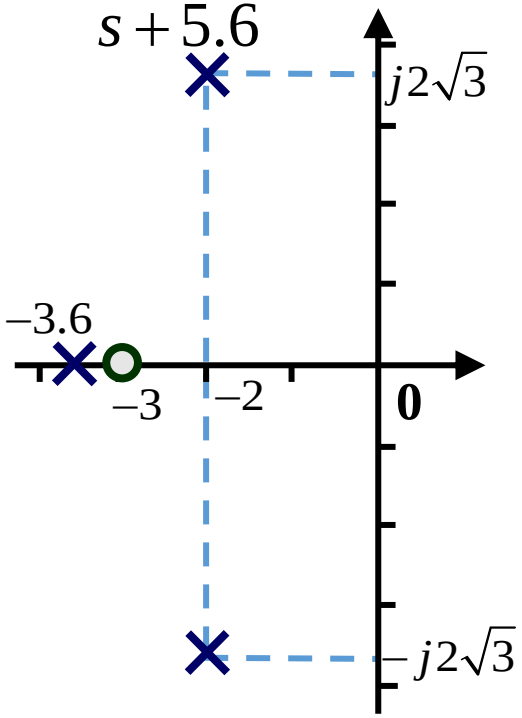
Lead Compensator Design



$$G_c(s) = 4.8\frac{s+3}{s+5.6}$$

Closed-loop TF $T(s) = \dfrac{19.2(s+3)}{(s+3.6)(s^2+4s+16)}$

The dominant poles $s_{1,2} = -2 \pm j2\sqrt{3}$ determine mostly the characteristics of the response.



They correspond to the desired poles.

The pole at $s = -3.6$ is farther from the jw-axis.

Therefore, its effect on the response dies out quickly.

Moreover, its effect is further reduced by the nearby zero at $s = -3$.

Lead Compensator Design

The dominant poles correspond to the desired poles and the specifications $\omega_n = 4,\ \zeta = 0.5,$ are met.

The above designs are two possible designs for compensator.

In the first design the zero of the compensator cancelled the pole of the plant and the resultant system was second order.

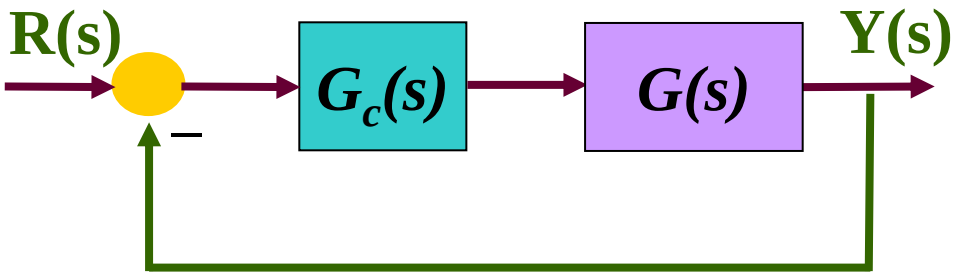
In the second design pole-zero cancellation did not occur and the resultant system was third order with dominant poles at the desired poles.

Note that, this is not necessarily the only design meeting all the specifications.

There may be many other configurations satisfying the requirements.

Lag Compensator Design Using Root-locus

The Lag Compensator



The TF of a phase lag compensator is given by

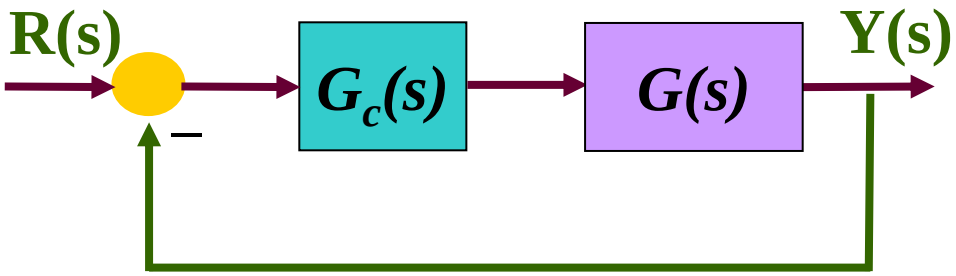
$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\beta\tau}} = K_c \frac{s+z}{s+p}, \beta > 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\beta\tau$$

Note that since $\beta > 1$ we have $-z = -1/\tau < -1/\beta\tau = -p$

The Lag Compensator



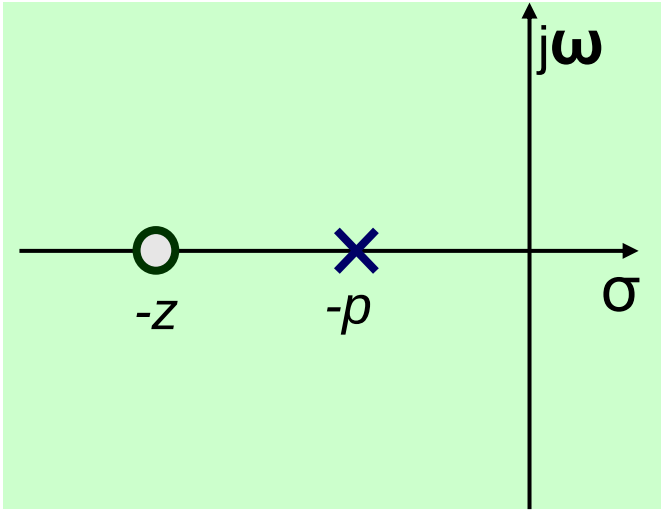
$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\beta\tau}} = K_c \frac{s+z}{s+p}, \beta > 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\beta\tau$$

$$-z = -1/\tau < -1/\beta\tau = -p$$

s-plane pole-zero configuration of a lag compensator



The Lag Compensator

Lag compensators are usually used for systems which exhibit satisfactory transient response characteristics, but unsatisfactory steady state characteristics.

Therefore, the compensator must increase the open-loop steady state gain without much altering the dominant pole locations (which determine the transient response characteristics).

To avoid large change in the dominant pole positions, the angle contribution of the compensator must be limited to a small amount, usually less than 5°.

This is achieved by placing the pole and the zero of the compensator relatively close to each other.

Also, by placing the pole and the zero of the compensator close to the origin it is possible to achieve large steady state gain.

The Lag Compensator

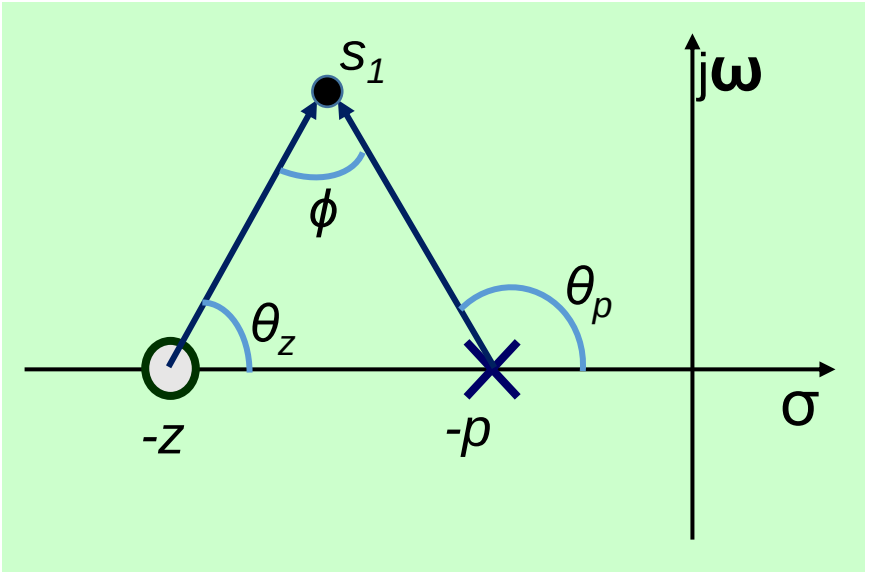
Let s_1 be one of the dominant poles of the original system.

At $s = s_1$ we have

$$G_c(s_1) = K_c \frac{s_1 + z}{s_1 + p}$$

Since the pole and the zero are very close to each other we have $|s_1 + z| \cong |s_1 + p|$ and

$$|G_c(s_1)| = K_c \frac{|s_1 + z|}{|s_1 + p|} \cong K_c$$



Similarly, the angle of $G_c(s_1)$ is

$$\left|\measuredangle\left(G_c(s_1)\right)\right| = \left|\measuredangle\left(s_1 + z\right) - \measuredangle\left(s_1 + p\right)\right| = \left|\theta_z - \theta_p\right| = \left|\phi\right| \le 5^o$$

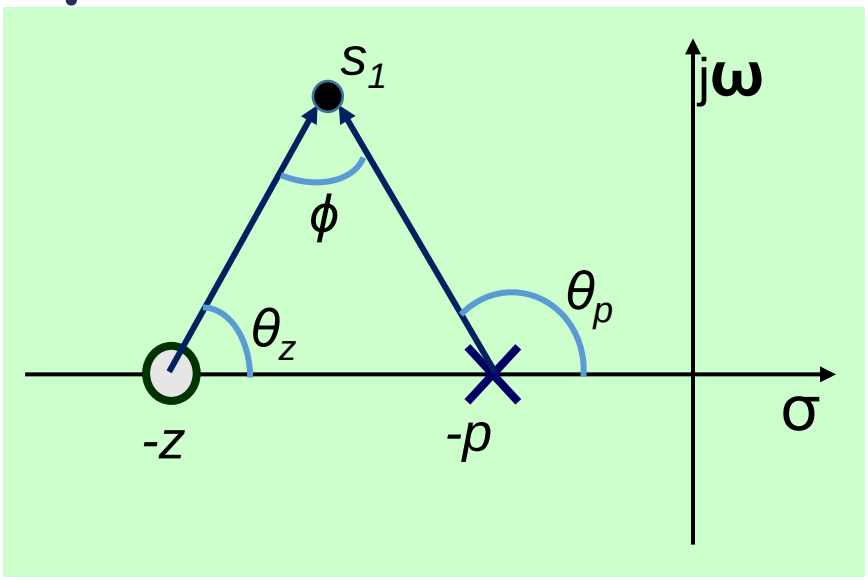
Therefore, if $K_c \cong 1$, then the transient response characteristics of the system will not be altered (much).

The Lag Compensator

The steady-state error characteristics of the system will be altered since

$$G_c(0) = \lim_{s\to 0} K_c \frac{s+z}{s+p} = K_c \frac{1/\tau}{1/\beta\tau} = \beta K_c$$

For $K_c \cong 1$ the open-loop gain is increased by a factor of $\beta > 1$



By choosing the zero and the pole of the compensator close to the origin we can achieve large β .

Consider the steady state velocity error constant $K_v = \lim_{s\to 0} sG(s)$

The velocity error constant of the compensated system is

$$\hat{K}_v = \lim_{s\to 0} sG_c(s)G(s) = \beta K_c K_v$$

The velocity constant is increased by a factor of β

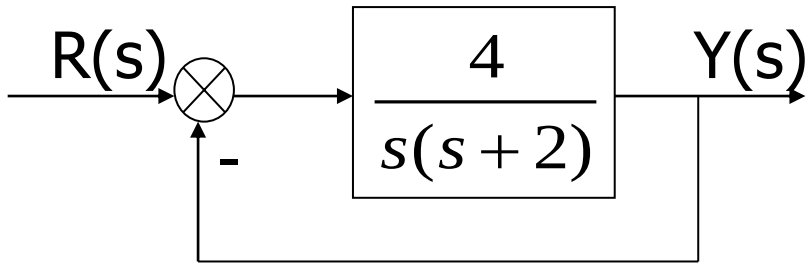
For $K_c \cong 1$ we have $\hat{K}_v = \beta K_v$.

Lag Compensator Design Procedure

1. Draw the root-locus of the uncompensated system and locate the dominant closed-loop poles which satisfy the transient response specifications.
2. Calculate the error constant for the uncompensated system and the amount of increase which is needed to satisfy the specifications.
3. Determine the pole and the zero of the compensator (place them close to each other and to the origin) that produce the necessary increase in the error constant without altering much the original root-locus. (If the angle of the compensator is kept small, there should not be a significant change in the root locus after adding the compensator.)
4. Draw the root-locus of the compensated system and locate the desired pole locations. Also, from the magnitude condition find (adjust) the compensator gain K_c corresponding to these pole locations.

Lag Compensator Design

Example: Consider again the system



Open-loop TF: $G(s) = \dfrac{4}{s(s+2)}$

Closed-loop TF: $T(s) = \dfrac{4}{s^2 + 2s + 4}$

For the current system we have: $\omega_n = 2,\ \zeta = 0.5,\ s_{1,2} = -1 \pm j\sqrt{3}$

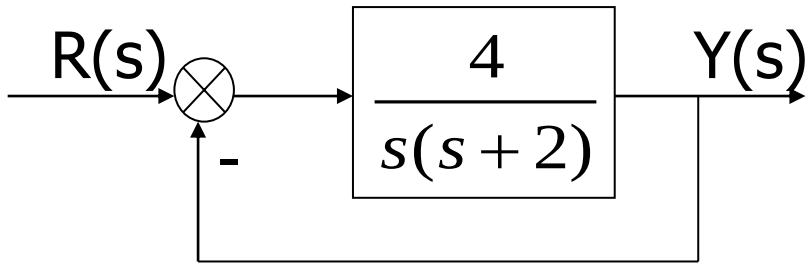
Design specifications:

Assume that $\omega_n = 2,\ \zeta = 0.5$ provide satisfactory transient response. However, it is required that the steady state error for a ramp input be less than 0.2.

We need $e_{ss} = \dfrac{1}{\hat{K}_v} < 0.2 \Rightarrow \quad \hat{K}_v > 5$ to be satisfied.

Lag Compensator Design

Example: Consider again the system



Open-loop TF: $G(s)=\dfrac{4}{s(s+2)}$

Closed-loop TF: $T(s)=\dfrac{4}{s^2+2s+4}$

For the current system we have: $\omega_n=2,\ \zeta=0.5,\ s_{1,2}=-1\pm j\sqrt{3}$

The velocity error constant of the uncompensated system is

$$K_v=\lim_{s\to 0}sG(s)=\lim_{s\to 0}s\frac{4}{s(s+2)}=2<5$$

The steady state error for a ramp input is

$$\Rightarrow\quad e_{ss}=\frac{1}{K_v}=\frac{1}{2}=0.5>0.2$$

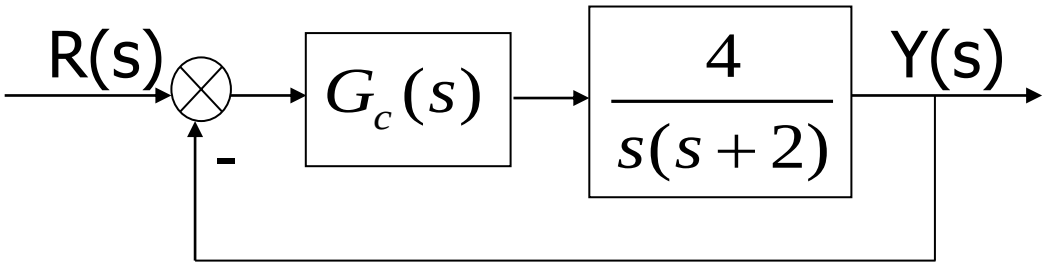
The ratio of the desired $\hat{K}_v$ and the actual K_v is

$$\frac{\hat{K}_v}{K_v}=\frac{5}{2}=2.5$$

$$\Rightarrow\quad \beta\geq 2.5$$ is needed assuming $K_c=1$

Lag Compensator Design

Use a lag compensator



$$G_c(s) = K_c \frac{s + 1/\tau}{s + 1/\beta\tau} = K_c \frac{s+z}{s+p}, \qquad \beta > 1$$

Let $K_c = 1$ and $\beta = 2.5$

Choose $z = 0.1$ and $p = 0.04$ for the compensator resulting in $\Rightarrow$ $G_c(s) = \dfrac{s+0.1}{s+0.04}$

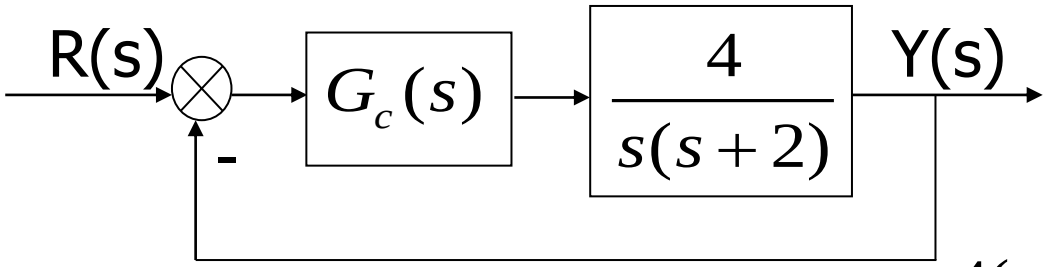
The angle of $G_c(s)$ at the (desired) pole $s_1 = -1 + j\sqrt{3}$ is

$$\measuredangle\left(G_c(s)\right)\Big|_{s=-1+j\sqrt{3}} = \measuredangle\left(-1+j\sqrt{3}+0.1\right) - \measuredangle\left(-1+j\sqrt{3}+0.04\right)$$

$$\cong 117^o - 119^o = -2^o$$

$\Rightarrow$ $-5^o < \measuredangle\left(G_c(s)\right) < 5^o$ is small (within desired limits)

Lag Compensator Design



$$G_c(s) = \frac{s+0.1}{s+0.04}$$

Open-loop TF $$G_c(s)G(s) = \frac{4(s+0.1)}{s(s+0.04)(s+2)}$$

Closed-loop TF

$$T(s) = \frac{4(s+0.1)}{s(s+0.04)(s+2)+4(s+0.1)} = \frac{4s+0.4}{s^3+2.04s^2+4.08s+0.4}$$

The CL poles $$s_{1,2} = -0.9685 \pm j1.7153,\ s_3 = -0.1031$$

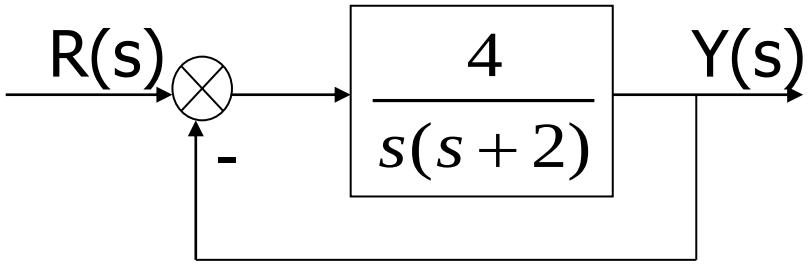
Note that, the new poles $s_{1,2}$ are very close to the original (desired) poles $s_{1,2} = -1 \pm j1.7321$.

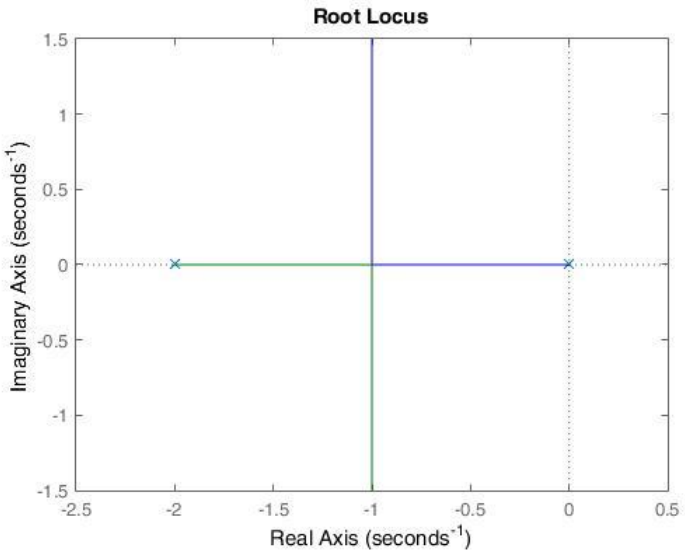
The new velocity error constant is

$$\hat{K}_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} s\frac{4(s+0.1)}{s(s+0.04)(s+2)} = \frac{4\times 0.1}{0.04\times 2} = 5$$

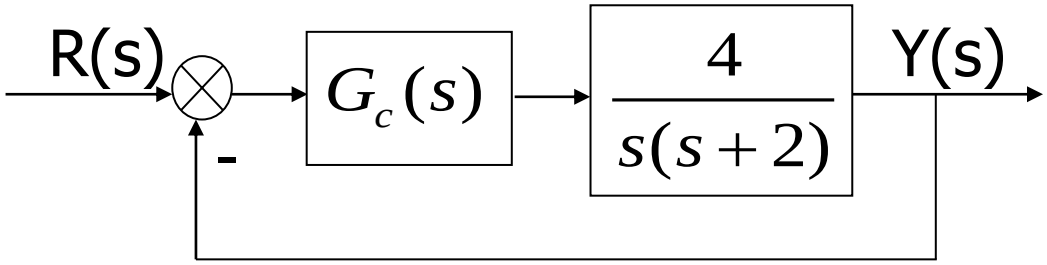
Lag Compensator Design

Root-locus of the original system

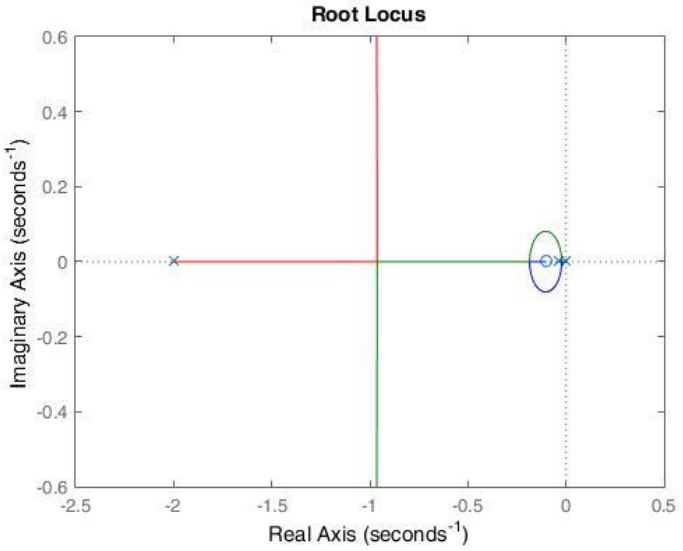




Root-locus of the compensated system

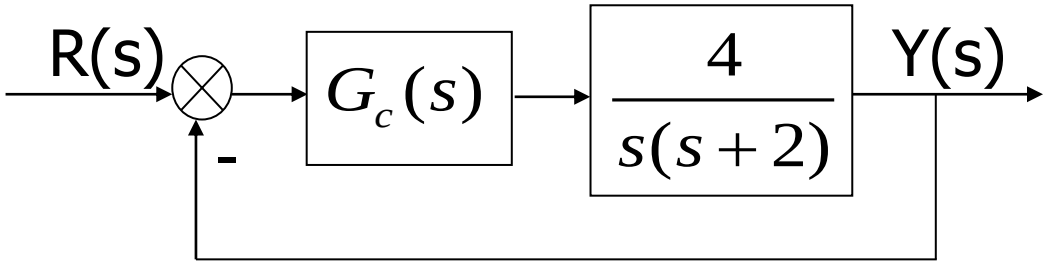


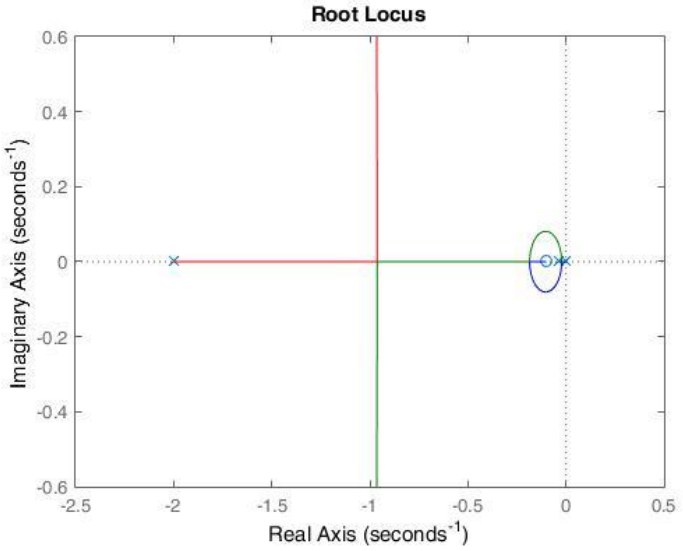
$$G_c(s)G(s) = \frac{4(s+0.1)}{s(s+0.04)(s+2)}$$



Lag Compensator Design

Root-locus of the compensated system





$$G_c(s)G(s) = \frac{4(s+0.1)}{s(s+0.04)(s+2)}$$

Centroid $\sigma_0 = \dfrac{0-0.04-2+0.1}{2} = -0.97$

Break-away/in points:

$$Z'(s)P(s) - Z(s)P'(s) = 0$$

$$s^3 + 1.17s^2 + 0.204s + 0.004 = 0$$

$$s_1 = -0.0224, s_2 = -0.1852,$$

$$s_3 = -0.9623$$

Note that the desired poles are

$$s_{1,2} = -1 \pm j1.7321$$

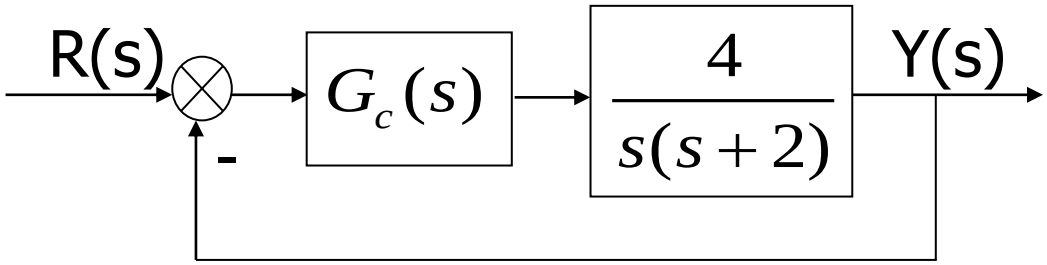
and we found the new poles

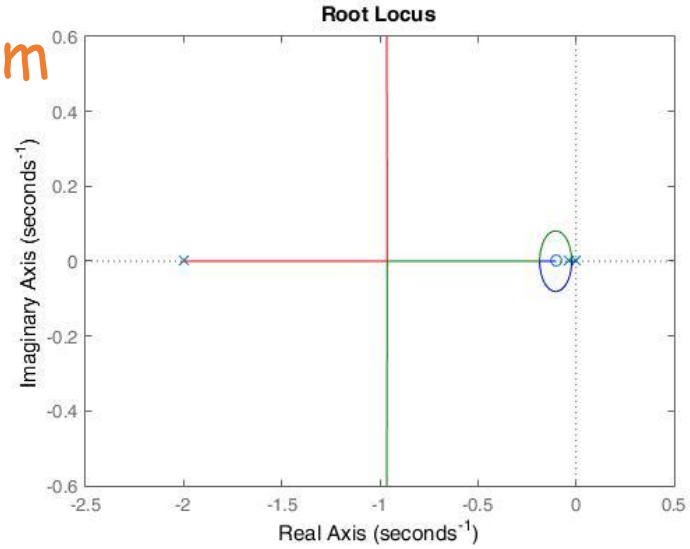
$$s_{1,2} = -0.9685 \pm j1.7153$$

satisfactory.

Lag Compensator Design

Root-locus of the compensated system





$$G_c(s)G(s) = \frac{4(s+0.1)}{s(s+0.04)(s+2)}$$

Instead of using $s_{1,2} = -0.9685 \pm j1.7153$ we could have chosen new poles from the root-locus of the compensated system (if the above poles were unsatisfactory or if we wanted poles closer to the desired ones).

For example, let us choose $s_{1,2} = -0.9685 \pm j1.7321$ which also belong to the root locus of the compensated system.

Then, we need to recalculate the controller gain K_c which will lead to these poles.

Lag Compensator Design

From the magnitude condition we have

$$\frac{4K_c\left|-0.9685+j1.7321+0.1\right|}{\left|-0.9685+j1.7321\right|\left|-0.9685+j1.7321+0.04\right|\left|-0.9685+j1.7321+2\right|}=1$$

$$\Rightarrow K_c=\frac{1.9845\times 1.9653\times 2.0160}{4\times 1.9376}=1.0145$$

For this gain we have

$$G_c(s)=1.0145\frac{s+0.1}{s+0.04} \quad \Rightarrow \quad G_c(s)G(s)=\frac{4.0580(s+0.1)}{s(s+0.04)(s+2)}$$

And the CL TF becomes

$$T(s)=\frac{4.058(s+0.1)}{(s+0.103)(s^2+1.937s+3.938)}$$

The CL poles

$$s_{1,2}=-0.9685\pm j1.7321,$$

$$s_3=-0.103$$

The new velocity error constant is

$$\hat{K}_v=\lim_{s\to 0}sG_c(s)G(s)=\lim_{s\to 0}s\frac{4.058(s+0.1)}{s(s+0.04)(s+2)}=\frac{4.058\times 0.1}{0.04\times 2}=5.0725$$

Lead-Lag Compensator Design Using Root-locus

The Lead-Lag Compensator

Lead compensators speed up the response (i.e., improve transient response) and increase the relative stability of the system.

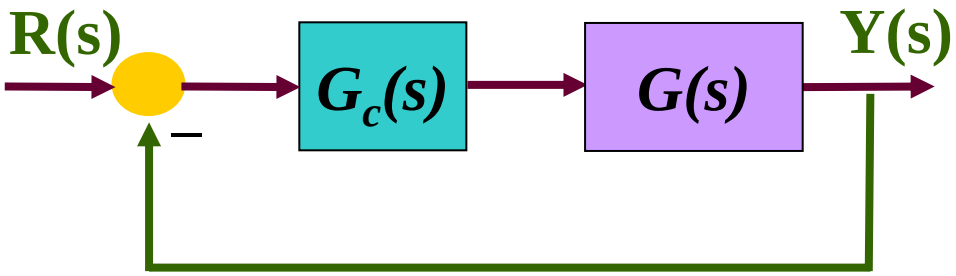
Lag compensators improve the steady-state accuracy but reduce the speed of the response.

If there is a need for improvement of both the transient response and the steady-state accuracy, then both lead compensator and lag compensator can be used simultaneously.

Instead of introducing the two compensators separately, it is better to combine them.

The resultant compensator is called lead-lag compensator.

The Lead-Lag Compensator



The TF of a lead-lag compensator is given by

$$G_c(s) = K_c \underbrace{\left(s+\frac{1}{\tau_1}\right)}_{Lead}\underbrace{\left(s+\frac{1}{\tau_2}\right)}_{Lag}\Big/\underbrace{\left(s+\frac{1}{\alpha\tau_1}\right)}_{Lead}\underbrace{\left(s+\frac{1}{\beta\tau_2}\right)}_{Lag}, \quad \begin{matrix} 0<\alpha<1 \\ \beta>1 \end{matrix}$$

We will consider two approaches for lead-lag compensator design.

Lead-Lag Compensator Design Procedure 1

This procedure is for the case in which $\beta \neq 1/\alpha$.

The procedure is a direct combination of the design of lead and the lag compensators.

The compensator has five parameters to be determined.

Procedure 1: Case $\beta \neq 1/\alpha$

1. From the desired performance specifications determine the desired locations for the dominant closed-loop poles.
2. Determine the angle deficiency ϕ of G(s) at the desired dominant pole locations. The phase-lead portion of the compensator must contribute that angle.

Lead-Lag Compensator Design Procedure 1

3. Determine the values of α, τ_1, K_c from the requirement that

$$\angle\left(\frac{s+1/\tau_1}{s+1/\alpha\tau_1}\right)\Bigg|_{s=s_1=\text{desired pole}} = \phi = \text{the angle defficiency of } G(s_1)$$

The angle condition

and

$$K_c\left|\frac{s+1/\tau_1}{s+1/\alpha\tau_1}G(s)\right|_{s=s_1=\text{desired pole}} = 1$$

The magnitude condition

Lead-Lag Compensator Design Procedure 1

4. From the specification for the steady state error (such as the velocity error constant $\hat{K}_v$) determine the value of β which will satisfy the requirements.

$$\hat{K}_v = \lim_{s\to 0} sG_c(s)G(s) = K_c\alpha\beta K_v$$

$\hat{K}_v$ - Error constant of compensated system

K_v - Error constant of uncompensated system

4. Choose τ_2 such that

$$\left|\frac{s+1/\tau_2}{s+1/\beta\tau_2}\right|_{s=s_1=\text{desired pole}} \cong 1 \quad \text{and} \quad \left|\measuredangle\left(\frac{s+1/\tau_2}{s+1/\beta\tau_2}\right)_{s=s_1=\text{desired pole}}\right| < 5^o$$

Lead-Lag Compensator Design Procedure 2

This procedure is for the case in which $\beta = 1/\alpha$.

The compensator has four parameters to be determined.

The TF of the compensator can be written as

$$G_c(s) = K_c \frac{\left(s+\frac{1}{\tau_1}\right)\left(s+\frac{1}{\tau_2}\right)}{\left(s+\frac{\beta}{\tau_1}\right)\left(s+\frac{1}{\beta\tau_2}\right)}, \beta > 1$$

Lead-Lag Compensator Design Procedure 2

Procedure 2: Case $\beta = 1/\alpha$

1. From the desired performance specifications determine the desired locations for the dominant closed-loop poles.

2. Determine the value of K_c from the specification for the steady state error (such as the velocity error constant $\hat{K}_v$)

$$\hat{K}_v = \lim_{s \to 0} sG_c(s)G(s) = K_c K_v$$

$\hat{K}_v$ - Error constant of compensated system

K_v - Error constant of uncompensated system

3. Determine the angle deficiency ϕ of G(s) at the desired dominant pole locations.

Lead-Lag Compensator Design Procedure 2

4. Determine the values of τ_1, β from the requirement that

$$\angle\left(\frac{s+1/\tau_1}{s+\beta/\tau_1}\right)\Bigg|_{s=s_1=\text{desired pole}} = \phi = \text{the angle defficiency of } G(s_1)$$

The angle condition

and

$$K_c\left|\frac{s+1/\tau_1}{s+\beta/\tau_1}G(s)\right|_{s=s_1=\text{desired pole}} = 1$$

The magnitude condition

Lead-Lag Compensator Design Procedure 2

4. Choose τ_2 sufficiently large so that

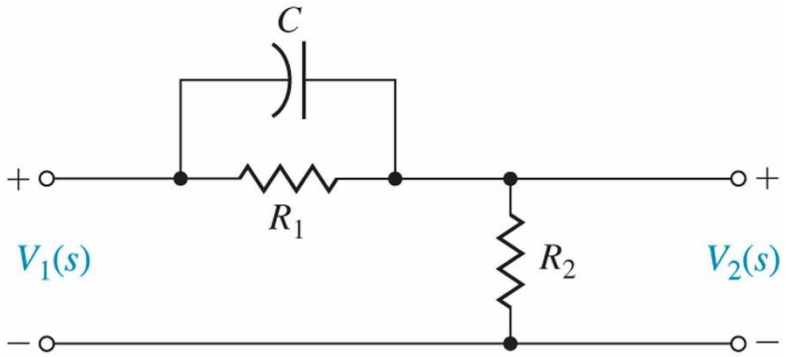
$$\left| \frac{s + 1/\tau_2}{s + 1/\beta\tau_2} \right|_{|s=s_1 = \text{desired pole}} \cong 1 \quad \text{and} \quad \left| \angle \left(\frac{s + 1/\tau_2}{s + 1/\beta\tau_2} \right)_{|s=s_1 = \text{desired pole}} \right| < 5^o$$

Note that the lead-lag compensator is a second order system and increases the order of the CL system by two. (Assuming that there are no pole-zero cancellations.)

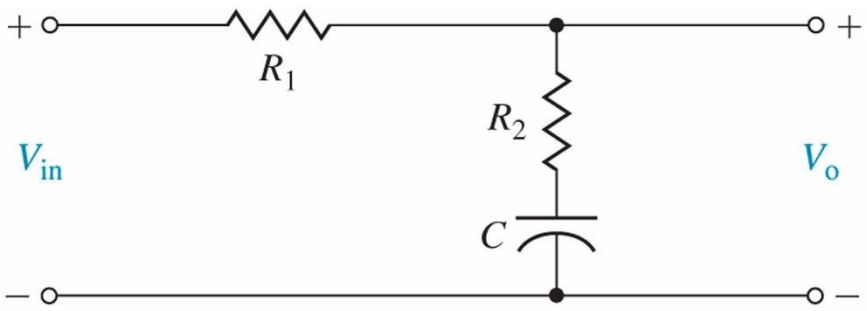
Please read the lead-lag compensator design examples from the book!

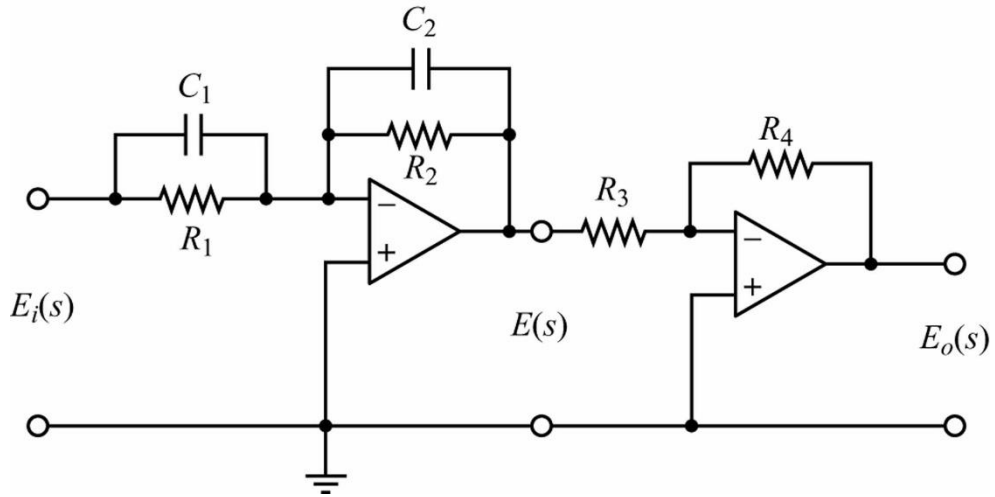
Possible Analog Implementation of Compensators

Phase-lead compensator



Phase-lag compensator





if $R_1 C_1 > R_2 C_2$
lead compensator

if $R_1 C_1 < R_2 C_2$
lag compensator

Possible Analog Implementation of Compensators

Lead-lag compensator

